

Binary Search

Session 7

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Outline

- 1 Beats and Bytes
- 2 Intuition
- 3 Analysis

Beats and Bytes

“Hard work builds strength. Smart work builds direction.”

Intuition

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Would you search in different ways in both books?

What if the book has 5 pages?

What if the book has 1000 pages?

In search of Mike Smith
Binary Search Animation

Analysis

If an array has **1,000,000 elements**, a linear search may take up to **1,000,000 comparisons** in the worst case.

How many comparisons do you think **binary search** would need?
(Most people guess a few thousand)



Figure: Akbar–Birbal and the Chessboard Grains

18,446,744,073,709,551,616 grains

- 450 TRILLION KG
- 7 CENTURIES of rice production of the world
- about 18 BILLION truckloads
- That's many times heavier than Mount Everest